

Elimination Half-lives of Polychlorinated Biphenyl Congeners in Children



The elimination kinetics of polychlorinated biphenyls (PCBs) in humans is difficult to assess in observational studies, because PCB exposure is never completely abolished. In a community with high dietary PCB exposures from whale blubber, we examined two groups of children with increased body burdens from breast-feeding. Follow-up was from ages 4.5 years to 7.5 years (99 subjects) and 7 to 14 years (101 subjects). The calculations were performed by the use of structural equation models, with adjustment for body weight and dietary blubber intake as the main source of postnatal exposure. As a likely result of background exposures, apparent elimination half-lives were unexpectedly long when based on results from all cohort members. Subjects with exposures above the median and in the highest quartile showed half-lives of about 3-4 years for CB-138, and 4.5-5.5 years for CB-105 and CB-118; 6.5-7.5 years for CB-156, CB-170, and CB-187; and 7-9 years for CB-153 and CB-180. The longest half-lives correspond to elimination of the parent PCB solely with a daily fat excretion rate of 1-2 g, while shorter half-lives assume metabolic break-down.